All formulae

1) $Simple\ linear\ regression:\ y = b_0 + b_1 * x_1$
2) $Multiple\ linear\ regression:\ y = b_0 + b_1 * x_1 + b_2 * x_2 + \ldots + b_n * x_n$
3) $error:\ \left(y_a - y_p\right) = Residual$
4) $SSE:\ \sum_{i=1}^{n}\left(y_{a_i} - y_{p_i}\right)^2 = RSE$
5) $MSE:\ \frac{1}{N} * \sum_{i=1}^{n}\left(y_{a_i} - y_{p_i}\right)^2$
6) $RMSE:\ \sqrt{\frac{1}{N} * \sum_{i=1}^{n}\left(y_{a_i} - y_{p_i}\right)^2}$
7) $Loss:\ \left(y_a - y_p\right)$
8) $Cost: j = \frac{1}{N} * \sum_{i=1}^{n}\left(y_{a_i} - y_{p_i}\right)^2$
9) $b_o = \overline{y_a} - b_1 * \overline{x}$

$$b_1 = \frac{\sum_i^n\left(\overline{y_a} - y_{ai}\right) * \left(\overline{x} - x_i\right)}{\sum_i^n\left(\overline{x} - x_i\right)^2} = \frac{Cov(x,y)}{Var(x)}$$

10)
11) $Correlation(r) = \frac{Cov(x,y)}{\sigma_x * \sigma_y}$
12) $RSE = \sqrt{\frac{RSS}{n-p-1}} = \sqrt{\frac{1}{n-p-1}\sum_{i=1}^{n}\left(y_{actual} - y_{predicted}\right)^2}$
13) $R - squared = 1 - \frac{sum\ of\ squared\ residuals\ (SSR)}{Total\ sum\ of\ squares(SST)}$
14) $R - squared = 1 - \frac{\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2}{\sum_{i=1}^{n}\left(y_i - \overline{y}\right)^2}$
15) $R_{Adj} = 1 - \frac{\left[1 - R_{square}\right] * (N-1)}{N-p-1}$
16) $AIC:\ \ 2P - 2ln(L)$
17) $BIC:\ \ p * \ln ln\,(N) - 2\ln ln\,(L)$
18) $VIF = \frac{1}{1 - R_{Square}}$
19) $DW = \frac{\sum_{t=2}^{n}\left(e_t - e_{t-1}\right)^2}{\sum_{t=1}^{n} e_t^2}$
20) $Rige\ cost\ function = \frac{1}{n} * \sum_i^n\left(y_a - y_p\right)^2 + \ \lambda * \sum_i^n b_i^2$

21) $Lasso\ cost\ function = \frac{1}{n} * \sum_{i}^{n} \left(y_a - y_p\right)^2 + \ \lambda * \sum_{i}^{n} |b_i|$